Vidusha Sanidu

AI/ML RESEARCHER & COMPUTER SCIENCE GRADUATE

Phone: +94 76 041 6256 | **Email:** vidushsanidu07@gmail.com | **Portfolio:** viduzx.xanvia.tech |
GitHub: github.com/VidushaSanidu | **LinkedIn:** linkedin.com/in/vidushasanidu

RESEARCH PROFILE & INTERESTS

Decentralized Intelligence & Applied ML Systems

Computer Science graduate (BSc Hons) and Research Assistant with specialized research and engineering experience in **federated learning, distributed AI systems, and data-driven optimization**. Proven track record in architecting, implementing, and evaluating machine learning frameworks on longitudinal real world clinical and spatial-temporal datasets. Experienced in decentralized model training, personalized federated aggregation across heterogeneous client environments, and production-grade agentic AI pipelines. Seeking to advance foundational and applied research in privacy-preserving machine learning, distributed optimization, and resource-efficient intelligent systems.

Decentralized Intelligence | **Privacy-Preserving Machine Learning** | **ML in Healthcare & Decision Support** | **Distributed Optimization** | **Resource-Efficient AI** | **Multi-Agent Systems**

EDUCATION

University of Peradeniya, Sri Lanka — 2021 Jun – 2025 Dec
Bachelor of Science (Honors) in Computer Science — **Final GPA: 3.67 / 4.00**

ACADEMIC RESEARCH EXPERIENCE

Research Assistant | AI-Based Healthcare Resource Distribution — May 2026 – Present

University Research Council (URC), University of Peradeniya

Project: "Effective AI-Based Resource Distribution Management System in Sri Lankan Hospitals"

- Designed and evaluated an ensemble learning framework for predictive hemodialysis management, leveraging longitudinal real-world clinical records collected from the Renal Care Unit at Teaching Hospital Peradeniya.

Research Collaborator | Federated Vehicular AI over 5G/6G — Oct 2026 – Present

University of Peradeniya (Supervisor: Dr. Erunika Dayaratna) In Progress

Study: "Federated AI Framework for Speed Prediction in Semi-Automated Electric Vehicles over Public 5G/6G"

- Collaborating on the design and integration of the decentralized federated learning architecture to support privacy-preserving vehicular speed prediction across public 5G/6G communication networks.

Undergraduate Thesis Research | Federated Spatial-Temporal Learning — 2024 – 2025

Department of Statistics & Computer Science, University of Peradeniya Under Review: Ceylon Journal of Science (CJS)

Thesis: "Personalized Pedestrian Trajectory Prediction using Federated AI" • Presented poster at Research Café 2026, University of Peradeniya

- Architected and implemented a custom **Light-SIAT** (Spatial Interaction Aware Temporal) deep learning model integrated with Federated Learning (**FedAvg / FedAtt**) to evaluate decentralized trajectory forecasting.
- Investigated the impact of decentralized training and client configurations on predictive performance, local updates, and the trade-off between model utility and data decentralization.
- Studied federated model aggregation and personalization approaches for improving performance across heterogeneous clients and distributed data environments.

Research Project | Domain Adaptation in Transit Systems — 2024

University of Peradeniya (Advisor: Dr. Samal Dharmarathne)

Project: "Domain Adaptation & Local Vehicle Detection in Sri Lankan Transit"

- Investigated domain-shift vulnerabilities in globally pre-trained object detection models when deployed on region specific vehicle morphologies (e.g., three-wheelers and localized transit buses).
- Curated and annotated a multi-condition local transit dataset across varied illumination and shadow profiles; applied selective layer-freezing and transfer learning on **YOLOv8** to substantially improve low-light detection mAP.

INDUSTRY & PROFESSIONAL EXPERIENCE

AI & Automation Engineer (Internship) **Sep 2025 – Jan 2026**

Taino Spain SL — Remote (Spain)

- Engineered production grade Retrieval Augmented Generation (RAG) architectures utilizing LLM backends and vector DBs.
- Resolved critical deployment bottlenecks by implementing automated hallucination filtering, token rate limit mitigation, context window compression, and custom deterministic tool calling interfaces.

SELECTED AI & SYSTEMS ENGINEERING PROJECTS

Applied Architectures & MLOps

Agentic Academic Assistant & Multi-Tool RAG Pipeline **Lead Developer**

Architecture & Stack: Python, FastAPI, LangChain, ChromaDB, Gemini API, OCR Modules, WebSockets, React.js

- Designed an interactive, multi-agent LLM system that automates personalized academic degree path generation by cross-referencing student transcripts, state pathway mandates, and institutional course catalogs.
- Orchestrated a dual agent routing workflow comprising a *General Q&A Agent* and a *Study Plan Generator Agent* via LangChain tools, dynamically dispatching queries between ChromaDB vector stores and real time GPA/prerequisite computation modules.
- Integrated automated OCR ingestion pipelines for unstructured transcript extraction alongside strict production guardrails, including semantic hallucination checks, context window optimization, and API rate limiting.

Echo-IC: Distributed Voice Synthesis & Acoustic Analysis Pipeline **ML Systems Engineer**

Architecture & Stack: PyTorch, XTTS Model, OpenAI Whisper API, Python, FastAPI, Redis, Docker, CI/CD

- Developed an end-to-end multimodal speech processing pipeline combining automated speech-to-text transcription (Whisper), LLM-driven semantic and paralinguistic voice analysis, and zero shot neural voice cloning (XTTS).
- Decoupled heavy GPU inference workloads using asynchronous Redis job queues and FastAPI microservices.

TECHNICAL SKILLS & CORE COMPETENCIES

Machine Learning & Deep Learning	PyTorch, TensorFlow, Scikit-learn, XGBoost, LightGBM, Random Forest, YOLO, Spatial-Temporal Modeling, Transfer Learning, Domain Adaptation, Model Fine-Tuning
Distributed & Decentralized AI	FedAvg, FedAtt, Decentralized & Client-Based Training, FlowerAI, Personalization
LLM Systems & GenAI	RAG, LangChain, ChromaDB, Prompt & Context Engineering, Whisper ASR, XTTS Voice Synthesis, OCR Pipelines
Languages, Cloud & MLOps	Python, C/C++, Java, TypeScript, JavaScript, SQL, LaTeX • FastAPI, NestJS, Node.js, Docker, Redis, Microservices, AWS, Linux, Git CI/CD

LEADERSHIP, SERVICE & COMPETITIVE HONORS

ACADEMIC LEADERSHIP & SERVICE

- **Vice President (2024 – 2025):** Computer Society, University of Peradeniya (CSUP). Spearheaded technical workshops, entertainment events, and industry-academia guest lecture series.
- **Webmaster (2024 – 2025):** IEEE Computer Society Student Branch Chapter, University of Peradeniya.

HACKATHONS & COMPETITIVE AWARDS

- **Winner (1st Place):** KickStart Weekend Boot Camp (2022), Career Guidance Unit (CGU), University of Peradeniya.
- **6th Place National Finalist:** Manthra CTF Cybersecurity Competition, IEEE CS NSBM Chapter.
- **Top 15 Finalist:** Mini Hackathon in Data Science, Statistics Circle, University of Colombo.

ACADEMIC REFERENCES

Prof. Saluka Kodituwakku
Senior Professor
Department of Statistics and Computer Science
University of Peradeniya, Sri Lanka
Relationship: Undergraduate Thesis & Research Supervisor

Dr. S. Maheswaran
Dean
Faculty of Management
University of Peradeniya, Sri Lanka
Relationship: Principal Investigator (URC Renal Care Project)